

Upper limb movements as digital biomarkers in people with ALS



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Summary

Background Objective evaluation of people with amyotrophic lateral sclerosis (PALS) in free-living settings is challenging. The introduction of portable digital devices, such as wearables and smartphones, may improve quantifying disease progression and hasten therapeutic development. However, there is a need for tools to characterize upper limb movements in neurologic disease and disability.

Methods Twenty PALS wore a wearable accelerometer, ActiGraph Insight Watch, on their wrist for six months. They also used Beiwe, a smartphone application that collected self-entry ALS Functional Rating Scale—Revised (ALSFRS-RSE) survey responses every 1–4 weeks. We developed several measures that quantify count and duration of upper limb movements: flexion, extension, supination, and pronation. New measures were compared against ALSFRS-RSE total score (Q1–12), and individual responses to specific questions related to handwriting (Q4), cutting food (Q5), dressing and performing hygiene (Q6), and turning in bed and adjusting bed clothes (Q7). Additional analysis considered adjusting for total activity counts (TAC).

Findings At baseline, PALS with higher Q1–12 performed more upper limb movements, and these movements were faster compared to individuals with more advanced disease. Most upper limb movement metrics had statistically significant change over time, indicating declining function either by decreasing count metrics or by increasing duration metric. All count and duration metrics were significantly associated with Q1–12, flexion and extension counts were significantly associated with Q6 and Q7, supination and pronation counts were also associated with Q4. All duration metrics were associated with Q6 and Q7. All duration metrics retained their statistical significance after adjusting for TAC.

Interpretation Wearable accelerometer data can be used to generate digital biomarkers on upper limb movements and facilitate patient monitoring in free-living environments. The presented method offers interpretable monitoring of patients' functioning and versatile tracking of disease progression in the limb of interest.

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Introduction

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disorder characterized primarily by loss of motor neurons resulting in progressive muscle weakness ultimately culminating in death.^{1–4} Current treatments slow but do not stop progression. New approaches to quantifying function may act as improved clinical trial outcome measures and hasten therapeutic development. Staff-administered outcome

measures, such as the ALS Functional Rating Scale-Revised (ALSFRS-R) and spirometry, remain the primary ALS clinical trial endpoints, though new scales are being developed.⁵ Although these tools provide non-invasive and diverse insights into the functional abilities of people with ALS (PALS), there are limitations, such as the requirement for frequent in-clinic visits and nonuniform gross motor function rating.^{6,7} Moreover, they suffer from recall bias and inter-rater

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Research in context

Evidence before this study

Wearable devices have a potential for remote monitoring and functional assessment in people with ALS. We searched PubMed for studies on wearable accelerometers and ALS with no language restrictions using combination of words or terms “wearable” OR “accelerometer” OR “accelerometry” AND “amyotrophic lateral sclerosis” OR “ALS”. We reviewed studies published before March 13, 2023. Previous studies predominantly assessed the relationship between progressing disability and overall activity. More granular investigations focused on gait and speech, while upper limb functioning in free-living setting was studied in two articles.

Added value of this study

Our study introduces a measure for quantifying upper limb functioning in people with ALS. The proposed metrics are highly correlated with self-assessed ALSFRS-R score as well as responses to individual questions on arm functioning. The method is open-source and can be used to analyze data collected with various wrist-worn devices that collect raw, sub-second level accelerometer data.

Implications of all the available evidence

Accelerometry-derived metrics track ALS disease progression over time and can serve as outcome measures in ALS clinical trials, and possibly other neurological diseases affecting arm function.

variability, hindering objective and quantitative symptom evaluation in trials.⁸

Wearable devices are increasingly used to study physical activity in populations with cardiovascular disease, multiple sclerosis, arthritis and other diseases.^{9–12} They have been successfully employed to study individuals with ALS, demonstrating the association between ALS progression and patterns of behavior and function in PALS, as measured by digital wearable devices, including total activity volume, active vs. sedentary time, and time spent at home.^{13–17} These objectively measured endpoints, can be collected essentially continuously and complement in-clinic assessments.¹⁸

While there is growing interest in overall activity obtained from wearable devices, less attention has been given to characterizing specific arm movements or activity in the context of neurological disorders and disability. The study of arm movement in ALS is clinically important given the substantial number of patients affected and degree of disability resulting from arm weakness, and could be an important ALS trial outcome measure, since the fine motor subdomain of the ALSFRS-R has been shown to change most rapidly,¹⁹ potentially providing good statistical power. At the same time, previous analyses may have been hampered by variability in volume and type of arm movements among individuals.²⁰

In this study, we introduce an algorithm that generates disease monitoring digital biomarkers using a single, wrist-worn, tri-axial accelerometer to identify and quantify upper limb movements in PALS. Digital quantification of forearm movements was accomplished by using the gravitational component of the raw acceleration data to ascertain changes in sensor orientation and determine the four fundamental upper limb movements: flexion, extension, supination, and pronation. Finally, movements were summarized in terms of their frequency and duration.

We evaluated the above metrics' utility for disease monitoring by (1) tracking the proposed digital biomarkers over time, (2) comparing them with smartphone-based self-entry ALSFRS-R (ALSFRS-RSE) scores,²¹ and (3) analyzing them against widely used total activity counts (TAC).²²

Methods

Cohort description

Population

Between January and December 2021, twenty-three PALS meeting the revised El-Escorial criteria²³ were remotely recruited, enrolled, and followed for 6 months. Recruitment took place at the Neurological Clinical Research Institute and Sean M. Healey & AMG Center for ALS at Massachusetts General Hospital. Other inclusion criteria considered age (18 years or older), smartphone ownership (running Android or iOS operating systems), and ability to provide informed consent, comply with study procedures, and ambulate at enrollment. Participants were included in the study regardless of sex or gender. Sex at birth was collected for the purpose of the study based on participant's self-report. All participants provided informed consent.

Ethics

All study procedures were approved by the Institutional Review Board (IRB) at Massachusetts General Brigham (MGB) and MGH Information Security Office (Protocol #:2016P000885). All data collection and management adhered to MGB, state, and national guidelines and regulations. Participants provided informed consent prior to their engagement in study procedures. Participants received \$50 every three months for continued participation.

Data collection

At baseline, study participants were asked to download and install on their smartphones the Beiwe application, which is the front-end component of the open-source, high-throughput digital phenotyping platform.²⁴ The application collects passive data and can be configured to deliver surveys and audio recordings. Here, we used it to deliver the self-entry ALSFRS-R (ALSFRS-RSE) questionnaire at baseline and every 1–4 weeks thereafter as we have described previously.²¹

Participants were asked to wear the ActiGraph Insight Watch (ActiGraph, Pensacola, FL) on their wrist of choice continuously, except for recharging (required every few weeks). The devices were configured to collect raw sub-second level accelerometer data with a 32 Hz sampling rate and a dynamic range of ± 8 g (gravitational units). The collected de-identified data were securely transferred to the ActiGraph CentrePoint Data Hub via Bluetooth and subsequently to the CentrePoint cloud using cellular data.

Estimation of upper limb movement metrics using raw accelerometer data

In our approach, we estimate the upper limb movements by measuring the gravitational component of the tri-axial raw accelerometer data collected from a wrist. Let the measured tri-axial acceleration be $x(t) = (x_1(t), x_2(t), x_3(t))$, where $x_1(t)$ is measured in the horizontal right-left direction, $x_2(t)$ in the horizontal forward-backward direction, and $x_3(t)$ in the vertical direction by a wearable accelerometer placed on the wrist. The acceleration vectors are directed as shown in Fig. 1d. The signal consists of a mixture of a linear and gravitational acceleration, i.e., those corresponding to the device motion and spatial orientation of the device, respectively (Fig. 1a). To approximate gravitational acceleration, the signal is processed using a low-pass finite impulse response filter (a digital filter that removes portion of a signal above prespecified frequency) of order 8 and a cut-off frequency of 0.1 Hz (Fig. 1b)²⁵ and transformed into spherical coordinates, pitch $\vartheta(t)$ and roll $\varphi(t)$ (Fig. 1c), using the formulas:²⁶

$$\vartheta(t) = \arctan\left(-x_2(t) / \sqrt{x_1(t)^2 + x_3(t)^2}\right), \vartheta(t) \in [-90, 90], \quad \text{Eq. 1}$$

$$\varphi(t) = \arctan(x_1(t) / x_3(t)), \varphi(t) \in [-180, 180] \quad \text{Eq. 2}$$

To identify and characterize specific upper limb movements, we segmented the signal based on continuous monotonic changes of pitch $\vartheta(t)$ and roll $\varphi(t)$. Specifically, we identified limb flexions as signal fragments when $\vartheta(t)$ is monotonically increasing, limb extensions when $\vartheta(t)$ is monotonically decreasing, limb supinations when $\varphi(t)$ is monotonically increasing, and

limb pronations when $\varphi(t)$ is monotonically decreasing. Each labelled movement was therefore described by its start and end times and angle estimates. For example, a movement starting at $t_1 = 1.0$ s, $\vartheta(t_1) = -90$ degrees and ending at $t_2 = 1.5$ s, $\vartheta(t_2) = 90$ degrees indicates a 180-degree elbow extended arm flexion (i.e., arm straight with hand closest to the ground to a position with arm straight and hand as high as possible above the head) that took 0.5 s.

We aggregated the cumulative counts and durations of upper limb movements. We propose various metrics, denoted by $X_{y,z}$, where X represents a type of metric: C stands for total count and D for average duration. The first subscript y represents a type of movement: f for flexion, e for extension, s for supination, and p for pronation. The second subscript z represents a minimum change of angle in a single movement, regardless of its duration, expressed in degrees. For example, $C_{f,45}$ is the total count of limb flexions by at least 45° , while $D_{s,135}$ represents an average duration of limb supinations by at least 135° (Table 1). In addition, we defined metrics that combine flexion and extension, as well as supination and pronation. These metrics use a dual notation for y , e.g., $C_{fe,45}$ refers to the total count of limb flexions and extensions by at least 45° .

Given varying levels of impairment, count metrics used all recorded movements and duration used the 10 fastest upper limb movements for a given day (24-h period). We report all 36 movement count and duration metrics but focus on the 12 metrics with 45-degree angle thresholds here (see Supplementary Appendix for the 24 remaining daily metrics).

The proposed metrics were validated analytically using independent data collected from ten ALS patients observed in controlled conditions (see Supplementary Appendix).

Of note, the presented approach also enables efficient identification of wrist orientation with prolonged stand-still, exploration of various movement trajectories, as well as quantification of various spatiotemporal movement features. For example, Fig. 1e depicts one patient's left forearm mobility over 20 weeks. Solid black lines illustrate daily upper limb flexions by more than 90° , while gray dots indicate all temporal upper limb orientations recorded a given day (32 per second, 2,764,800 per day) and expressed by two angles, pitch and roll.

Statistics

The analysis considered data obtained in a prospective study. Sample size was determined by data availability from the completed study. As there was no treatment, there was no randomization or blinding.

After data preprocessing (see Supplementary Appendix), the statistical analysis sample included wearable device data from valid days, defined as days

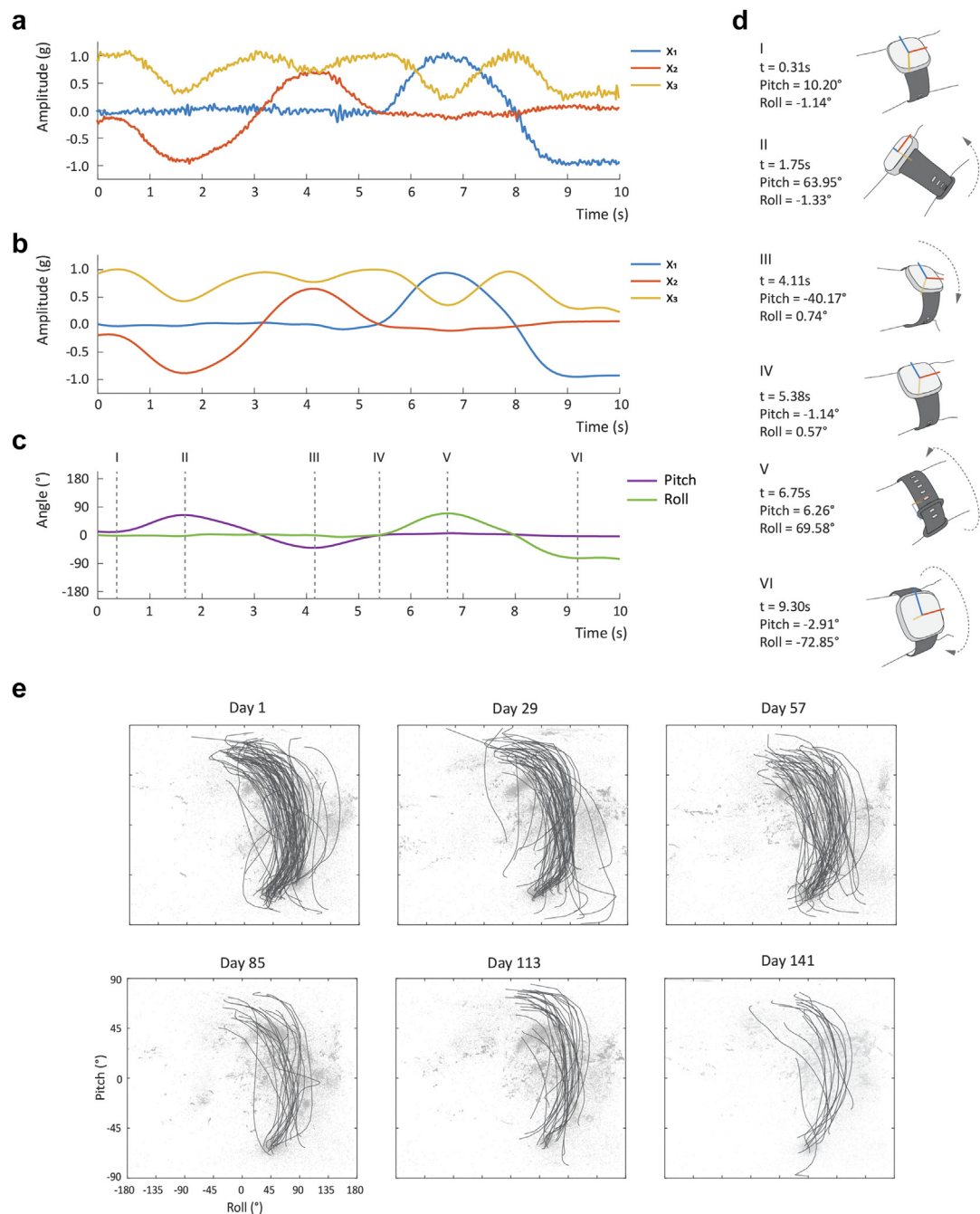


Fig. 1: Calculation, interpretation, and application of forearm movement detection method. Wearable accelerometers collect data in three orthogonal axes (a). The data are smoothed using a low-pass filter with cut-off frequency equal to 0.1 Hz (b). The filtered data are transformed into pitch and roll angles (c). The angles indicate temporal sensor (forearm) orientation, here visualized from the first-person view on the left wrist (d). Positions I-III depict forearm flexions and extensions, while positions IV-VI represent supination and pronation. The proposed algorithm recognizes movement between positions I and II as a forearm flexion by 53.75° that took 1.44s, while movement between II and III is accounted as a forearm extension by 104.12° (2.36s). Movements between position IV and V is recognized as a forearm supination by 69.01° (1.38s), while between position V and VI as a forearm pronation by 142.43° (2.55s). The device coordinate system was identified based on information provided in their technical manual. This method allows tracking changes in forearm activity over time. In this example (e), the raw real-life data are analyzed using one day of observation at day 1, 29, 57, 85, 113, and 141 (i.e., at baseline, after 4, 8, 12, 16, and 20 weeks). Each grey dot corresponds to one data point. Each black line corresponds to one forearm flexion by more than 90° . The patient performed 69, 52, 57, 20, 27, and 14 forearm flexions on the respective listed days, indicating a decline in upper limb function.

Proposed digital biomarker	Interpretation
$C_{f,45^\circ}$ $C_{f,90^\circ}$ $C_{f,135^\circ}$	Total daily count of flexions by at least 45, 90, and 135°
$C_{e,45^\circ}$ $C_{e,90^\circ}$ $C_{e,135^\circ}$	Total daily count of extensions by at least 45, 90, and 135°
$C_{s,45^\circ}$ $C_{s,90^\circ}$ $C_{s,135^\circ}$	Total daily count of supinations by at least 45, 90, and 135°
$C_{p,45^\circ}$ $C_{p,90^\circ}$ $C_{p,135^\circ}$	Total daily count of pronations by at least 45, 90, and 135°
$C_{fe,45^\circ}$ $C_{fe,90^\circ}$ $C_{fe,135^\circ}$	Total daily count of flexions and extensions by at least 45, 90, and 135°
$C_{sp,45^\circ}$ $C_{sp,90^\circ}$ $C_{sp,135^\circ}$	Total daily count of supinations and pronations by at least 45, 90, and 135°
$D_{f,45^\circ}$ $D_{f,90^\circ}$ $D_{f,135^\circ}$	Average daily duration of 10 fastest flexions by at least 45, 90, and 135°
$D_{e,45^\circ}$ $D_{e,90^\circ}$ $D_{e,135^\circ}$	Average daily duration of 10 fastest extensions by at least 45, 90, and 135°
$D_{s,45^\circ}$ $D_{s,90^\circ}$ $D_{s,135^\circ}$	Average daily duration of 10 fastest supinations by at least 45, 90, and 135°
$D_{p,45^\circ}$ $D_{p,90^\circ}$ $D_{p,135^\circ}$	Average daily duration of 10 fastest pronations by at least 45, 90, and 135°
$D_{fe,45^\circ}$ $D_{fe,90^\circ}$ $D_{fe,135^\circ}$	Average daily duration of 10 fastest flexions and extensions by at least 45, 90, and 135°
$D_{sp,45^\circ}$ $D_{sp,90^\circ}$ $D_{sp,135^\circ}$	Average daily duration of 10 fastest supinations and pronations by at least 45, 90, and 135°

Table 1: Notation of the proposed upper limb movement digital biomarkers with angles listed in respective order of notation.

with at least 21 h (1260 min) of cumulative wear-time as determined by the algorithm of Choi and colleagues,²⁷ and survey scores from participants with at least two complete ALSFRS-RSE surveys. The number of complete ALSFRS-RSE survey submissions and average time between the surveys were computed for each participant and characterized using mean, standard deviation (SD), median, and range. Wrist-worn sensor wear compliance was calculated as valid days divided by days in the observation period.

To estimate population-level baseline and monthly rates of change of ALSFRS-RSE scores, we fitted six linear mixed-effect models (LMMs). In each model, we specified time (months elapsed from the start) as a fixed effect and included patient-specific random intercept and random slope. The ALSFRS-RSE has 12-items/questions (Q), each scored 0–4 points with lower scores signifying more impairment. We used the following outcomes based on ALSFRS-RSE scores: summed total score (denoted Q1–12),²⁸ summed scores for Q4–7 (Q4–7), and individual scores for Q4 (handwriting), Q5 (cutting food and handling utensils), Q6 (dressing and hygiene), and Q7 (turning in bed and adjusting bedclothes).

We calculated the Pearson correlation coefficient for each pair of baseline metrics: 12 upper limb movement metrics (each for 45°), total activity counts (TAC), Q1–12, Q4–7, Q4, Q5, Q6, and Q7. TAC was calculated as a daily (24 h) sum of minute-level activity counts from the data provided by the vendor.²⁹ For each of the limb movement metrics and TAC, we computed an average of a participant's daily metrics within ± 10 days from the baseline ALSFRS-RSE survey.

To estimate population-level baseline values and monthly rates of change, LMMs were fitted with the proposed limb movement metrics as the outcome, time (months elapsed from the start of observation) was included as a fixed effect, and we also included patient-specific random intercept and random slope. The LMM were also used to determine associations between the

proposed metrics and ALSFRS-RSE (see [Supplementary Appendix](#)).

When appropriate, we report 95% confidence intervals (CI) and p-values ($\alpha = 0.05$) for fixed effects estimates, and we report R-squared marginal (R2m; variance explained by fixed factors) and R-squared conditional (R2c; variance explained by both fixed and random factors) for LMM fits.^{30,31} The reported p-values were not adjusted for multiple testing (see [Supplementary Appendix](#)).

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Results

Demographics, compliance, ALSFRS-RSE status

Out of 23 enrolled participants, one was unable to download the application and two did not provide at least 2 ALSFRS-RSE surveys. In the final sample ($N = 20$, [Table 2](#)), there was a higher proportion of male ($N = 12$; 60%), white ($N = 17$; 85%), and non-Hispanic or Latino participants ($N = 18$; 90%). Most participants were right-handed ($N = 15$; 75%). Handedness was not collected for three participants (15%).

Participants completed a median of 6 surveys (range 2–15), and the median time elapsed between surveys was 32 days (range 6–112). We collected wrist-worn accelerometer data for a median of 178 days (range 23–191). Participants were highly compliant: median sensor wear-time was 124 valid days (range 3–165).

The estimated baseline Q1–12 score was 34.4 (95% CI: [30.4, 38.3]), and its monthly change was -0.76 (95% CI: $[-1.09, -0.42]$). Statistically significant decline over time was observed for 4 out of 6 outcomes: Q1–12,

Population	
Total number of participants, N	20
Demographics	
Age, mean (SD), median [min, max]	61.4 (10.6), 64 [34, 76]
Male, N (%)	12 (60%)
Female, N (%)	8 (40%)
Race	
White, N (%)	17 (85%)
More Than One Race, N (%)	2 (10%)
Native Hawaiian or Other Pacific Islander, N (%)	1 (5%)
Handedness	
Right	15 (75%)
Left	2 (10%)
Unknown	3 (15%)
Ethnicity	
Not Hispanic or Latino, N (%)	18 (90%)
Mobile operating system	
iOS, N (%)	15 (75%)
Android, N (%)	5 (25%)
Data collection compliance	
Smartphone data	
Number of surveys, mean (SD), median [min, max]	6.9 (3.6), 6 [2, 15]
Days between surveys, mean (SD), median [min, max]	36.8 (24.0), 32 [6, 112]
Accelerometer data	
Days with collected accelerometer data, mean (SD), median [min, max]	170.4 (35.6), 178 [23, 191]
Days with valid accelerometer data, mean (SD), median [min, max]	97.4 (51.9), 124 [3, 165]

For compliance, patient-specific values are aggregated across all study participants. A valid day in the observation period was defined as a day with at least 1260 device wear time minutes (21 h).

Table 2: Baseline demographics, ALSFRS-RSE survey and wearable device compliance.

Q4–7, Q6, and Q7. Average baseline and monthly change of ALSFRS-RSE scores are presented in [Table 3](#).

Upper limb movement metrics

Baseline correlations

At baseline, upper limb movement count metrics had moderate and positive correlation with the ALSFRS-RSE total score (Pearson $R \geq 0.54$), whereas upper limb movement duration metrics had a negative correlation ($R \leq -0.54$) ([Figure S1 in Supplementary Appendix](#)).³² PALS with higher overall levels of functioning on the ALSFRS-RSE performed more limb movements, and these movements were faster compared to individuals with more advanced disease.

The correlation between individual questions and the proposed metrics was moderate except for the flexion and extension duration metrics, which were highly negatively correlated with Q5 (≤ -0.75). Notably, negligible correlation was observed between the limb metrics and Q4–Q7 (range -0.24 to 0.36).

The correlation between TAC and the proposed metrics was very high and positive for count metrics (range 0.94 – 0.98) and high and negative for duration

metrics (range -0.78 to -0.81). (The results corresponding to metrics with angle thresholds of 90° and 135° are provided in [Supplementary Appendix](#)).

Baseline and monthly change

Linear mixed effects models (LMMs) were used to quantify average baseline values and monthly change of the proposed daily metrics ([Table 3](#)). Baseline estimates differed between movement types and were lower for daily upper limb flexion (581.4) and extension (595.3) counts compared to supination (1370) and pronation (1392) counts. At baseline, upper limb flexions and extensions took longer than supinations and pronations (0.704 s and 0.707 s vs. 0.473 s and 0.500 s, respectively). All metrics except duration of limb pronations demonstrated statistically significant change over time, indicating declining function either by decreasing daily count metrics or by increasing duration metrics. All metrics had very high R^2 (≥ 0.82). Patient-specific conditional means as well as population means for all limb movement daily measures are displayed in [Fig. 2](#). (For sex-based stratified data results, see [Supplementary Appendix](#)).

Association with ALSFRS-RSE scores

All daily upper limb measures were significantly associated with Q1–12 with a very high R^2 (all >0.96). The estimated slopes were positive for the count metrics and negative for the duration metrics, indicating a link between more frequent and faster limb movements and better limb function ([Fig. 3](#)). Metrics indicating frequency of daily flexions and extensions were also significantly associated with Q6 and Q7, while metrics on daily supinations and pronations were additionally associated with Q4. Only $C_{p,45}$ and $C_{sp,45}$ were significantly associated with Q5. All duration metrics were significantly associated with Q6 and Q7, but not with Q4 and Q5. (For sex-based stratified data results, see [Supplementary Appendix](#)).

Model outcomes provided in [Supplementary Appendix](#) allow for additional interpretation of these associations on a population level ([Table S2](#)). For example, a unit change of Q1–12 corresponded to an increase of about 141 daily upper limb flexions by at least 45° ($C_{f,45}$), 135 extensions ($C_{e,45}$), 247 supinations ($C_{s,45}$), and 215 pronations ($C_{p,45}$). The same change in Q1–12 corresponded to a decrease of about 69 ms (milliseconds) in average duration of upper limb flexions by at least 45° ($D_{f,45}$), 64 ms in duration of extensions ($D_{e,45}$), 59 ms in duration of supinations ($D_{s,45}$), and 51 ms in durations of pronations ($D_{p,45}$).

Association with ALSFRS-RSE scores while adjusting for TAC
Additional analysis was performed to examine whether the proposed upper limb movement metrics retain their statistical association with ALSFRS-RSE scores after adjusting for TAC. A comprehensive visualization of the

	Baseline estimate [95% CI]	Monthly change estimate [95% CI] (p-val.)	R2m	R2c
ALSFRS-RSE score				
Q1-12	34.37 [30.42, 38.32]	-0.757 [-1.090, -0.424] (0.000)	0.046	0.989
Q4-7	10.94 [9.222, 12.65]	-0.263 [-0.407, -0.120] (0.002)	0.029	0.990
Q4	2.912 [2.475, 3.350]	-0.054 [-0.110, 0.002] (0.058)	0.016	0.948
Q5	2.738 [2.196, 3.281]	-0.026 [-0.100, 0.047] (0.453)	0.003	0.941
Q6	2.478 [2.005, 2.951]	-0.066 [-0.109, -0.024] (0.005)	0.024	0.956
Q7	2.798 [2.270, 3.326]	-0.088 [-0.149, -0.026] (0.008)	0.044	0.951
Proposed digital biomarker				
$C_{f,45}$	581.4 [426.7, 736.0]	-28.30 [-46.28, -10.32] (0.004)	0.020	0.857
$C_{e,45}$	595.3 [435.0, 755.7]	-29.13 [-47.44, -10.82] (0.004)	0.020	0.862
$C_{fe,45}$	1177 [861.9, 1492]	-57.43 [-93.68, -21.18] (0.004)	0.020	0.860
$C_{s,45}$	1370 [1082, 1659]	-60.60 [-103.3, -17.91] (0.008)	0.026	0.833
$C_{p,45}$	1392 [1103, 1681]	-60.98 [-103.3, -18.68] (0.008)	0.026	0.832
$C_{sp,45}$	2762 [2185, 3339]	-121.6 [-206.5, -36.66] (0.008)	0.026	0.833
$D_{f,45}$	0.704 [0.593, 0.816]	0.012 [0.006, 0.018] (0.001)	0.006	0.924
$D_{e,45}$	0.707 [0.600, 0.814]	0.014 [0.008, 0.021] (0.000)	0.009	0.918
$D_{fe,45}$	0.706 [0.597, 0.814]	0.014 [0.007, 0.020] (0.000)	0.008	0.939
$D_{s,45}$	0.473 [0.406, 0.540]	0.005 [0.000, 0.010] (0.040)	0.003	0.874
$D_{p,45}$	0.500 [0.425, 0.576]	0.006 [0.000, 0.013] (0.064)	0.004	0.904
$D_{sp,45}$	0.487 [0.416, 0.558]	0.006 [0.000, 0.012] (0.043)	0.004	0.920

Note: CI, estimate's confidence interval obtained from LMM estimation; R2m, LMM marginal coefficient of determination; R2c, LMM conditional coefficient of determination.

Table 3: Average baseline and monthly change in ALSFRS-RSE scores and the proposed upper limb movement daily metrics.

LMM-estimated coefficient for upper limb movement metric covariates—for 6 different ALSFRS-RSE scores outcomes and 3 different sets of LMMs (varying by not adjusting and adjusting for TAC and inclusion of random slope)—is presented in Fig. 3.

Upper limb movement count metrics were not significantly associated with the ALSFRS-RSE when adjusting for TAC (except for Q6 and counts of supination and pronation). Importantly, flexion and extension duration metrics were significantly associated with 5 out of 6 ALSFRS-RSE outcomes after adjusting for TAC (Q1-12, Q4-7, Q4, Q5, and Q6); supination and pronation duration metrics retained their statistical significance after adjusting for TAC for 3 out of 6 ALSFRS-RSE outcomes (Q1-12, Q4-7, and Q6).

Discussion

Monitoring of arm function with digital quantitative tools, such as accelerometers, has been challenging because arm movements are substantially affected by physiological and habitual differences between individuals and differences in individuals' built environment.³³ As a result, upper limb activity monitoring has occurred predominantly in controlled settings with individuals performing a predefined set of activities. Under these conditions, researchers have proposed methods to recognize basic functional primitives (e.g., reaching, retrieving, idling) using heuristics (e.g., sequence of sensor orientations³⁴) or machine learning

techniques (e.g., *k*-mean clustering³⁵). Only recently have free-living wrist data been used to characterize neurological disease. For example, Gupta et al. investigated statistical features (mean and standard deviation of duration, distance, and peak velocity) of four movement groups (categorized based on duration and direction of movements) derived from velocity time series to study ataxia-telangiectasia.³⁶ Holdom et al. determined the association between wrist- and hip-based physical activity estimates (time spend active, signal variation in each axis, vector magnitude) with ALSFRS-R,¹⁷ while Vieira et al. predicted ALSFRS-R scores using convolutional neural networks trained with time- and frequency-domain features.¹⁸

In ALS, upper limb impairment can be assessed by specific questions of the ALSFRS-R, which score handwriting, cutting food and handling utensils or ability to manage feeding tube manipulation, managing dressing and hygiene and turning in bed (questions Q4-Q7, respectively). These specific items mostly assess fine rather than gross motor control, and more broadly the scale has several recognized limitations in detecting impairment resulting from hand dominance vs. the affected limb as well as multidimensionality.^{6,37,38} The available literature focused on different aspects of upper limb function and lack the ability to distinguish between different types of activity.^{14,17,39-41}

We describe a method for quantifying upper limb movements using wrist-worn accelerometers. This approach is unconstrained by initial sensor laterality

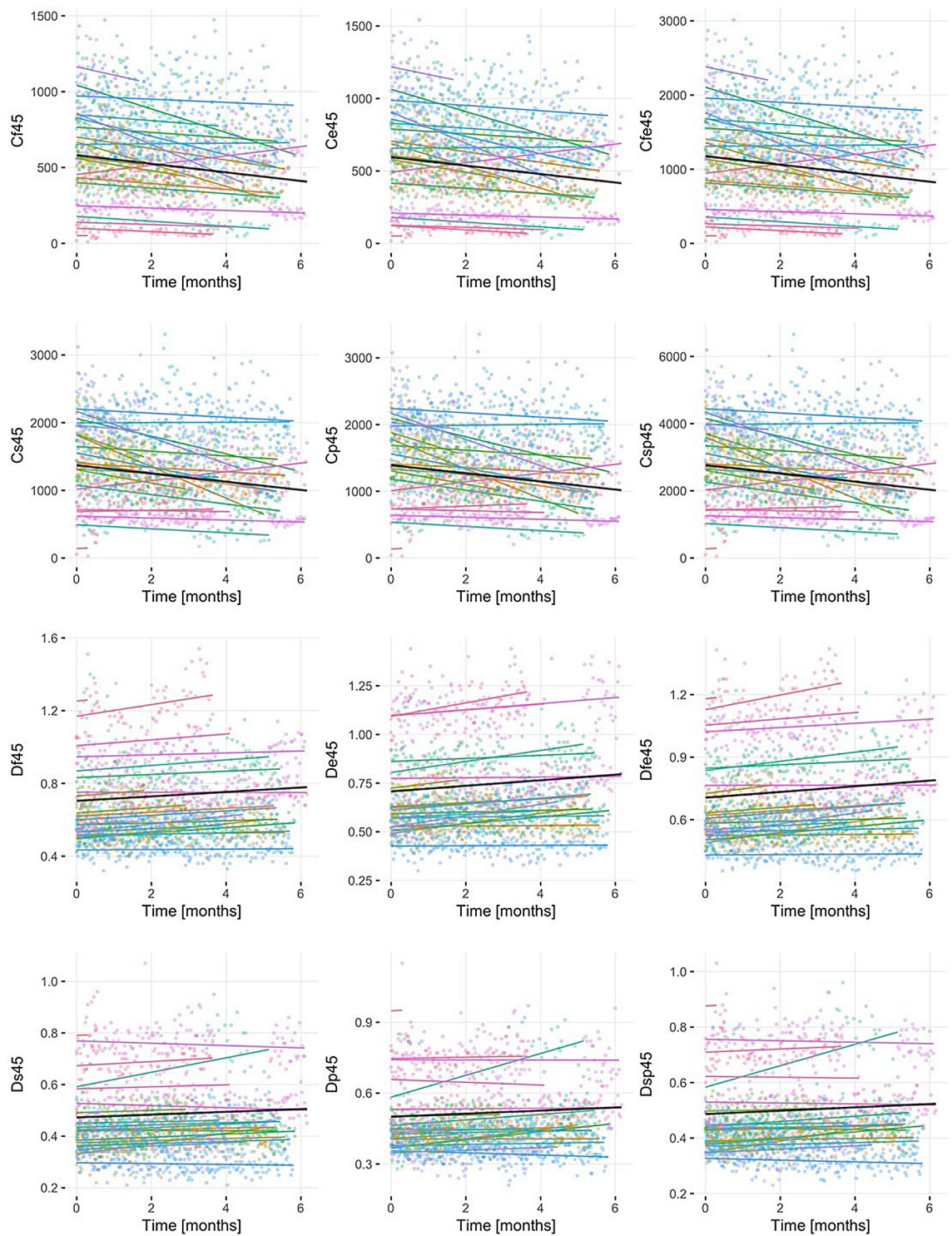


Fig. 2: Daily measure baseline and monthly change in the daily measure. In each plot, colored lines and points represent participant's LMM-estimated conditional mean values and observed data, respectively; the color scheme is maintained across plots. Black lines represent population mean values.

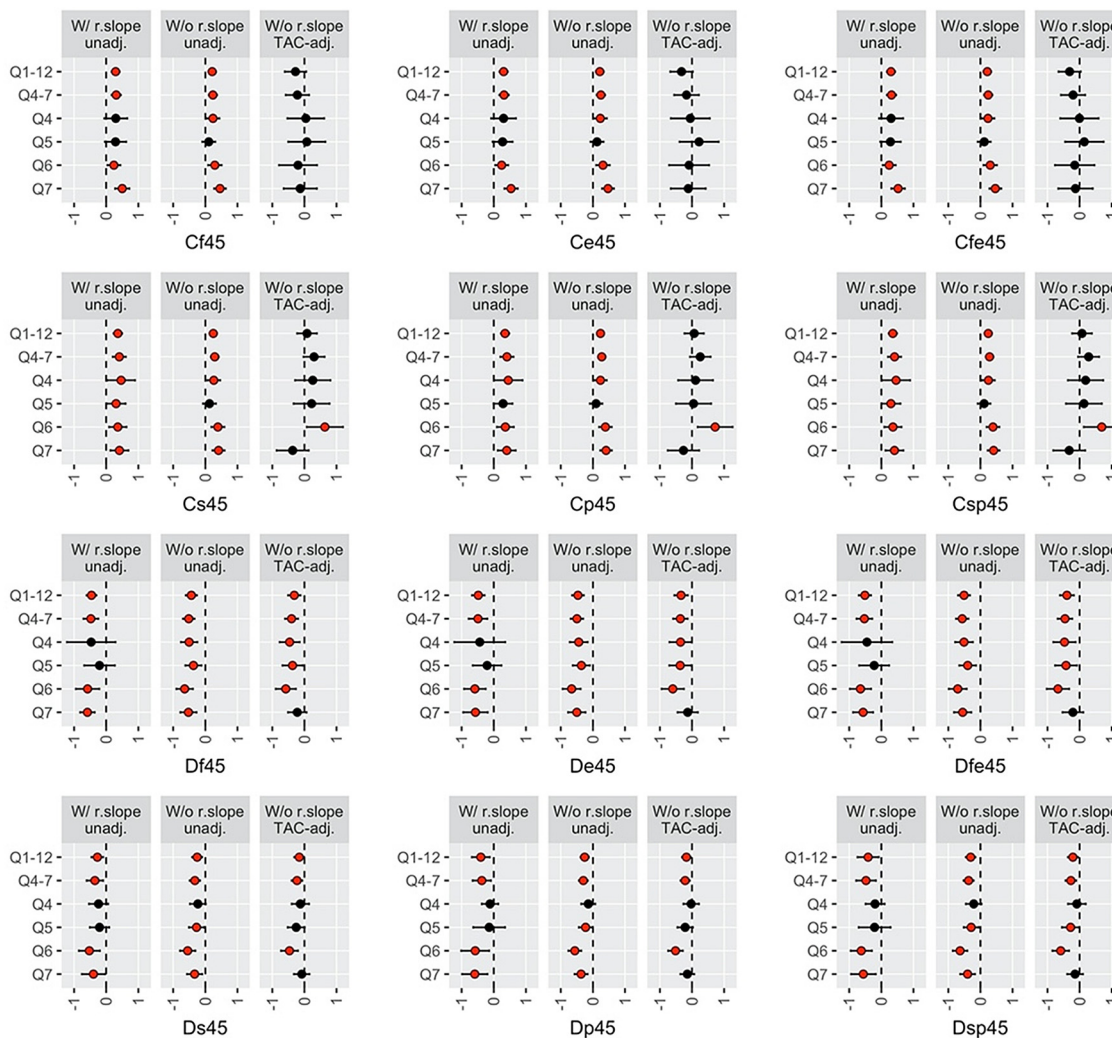


Fig. 3: Population-level effect coefficient estimate and 95% CI for the association between a daily measure (x-axis) and an ALSFRS-RSE-based outcome (y-axis). In each of the 12 plots, the three vertical subplot panels correspond to three sets of LMM models: (1) with patient-specific random intercept and random slope (left subplot panel—"W/ r.slope, unadj."), (2) with patient-specific random intercept only (middle subplot panel, "W/o r.slope, unadj."), (3) with patient-specific random intercept only and adjusted for a TAC covariate (right subplot panel—"W/o r.slope, TAC-adj."). In all LMM model estimation, both daily measures and the outcome were standardized to have a mean of 0 and a standard deviation of 1. Red points represent statistically significant associations (an effect for which 95% CI did not overlap with 0).

(left vs. right wrist), placement (top vs. bottom of the wrist), and orientation (anterior vs. posterior orientation), as long as the device's positioning is consistent during the 24-h wear-period. This is because upper limb movements and their temporal course are identified with reference to initial device orientation. From this data, we derived four fundamental movements: flexion, extension, pronation, and supination, and several additional upper limb movement metrics. The baseline and longitudinal "total daily count" and "average daily duration" of the 10 fastest movements of a given type are shown. These proposed metrics were compared against self-entry ALSFRS-R (ALSFRS-RSE)⁷ which has

demonstrated excellent correlation with staff-administered ALSFRS-R.²¹

Our approach has several strengths. First, our method is intuitive with respect to human kinesiology, quantifying upper limb metrics that are easy to conceptualize for direct monitoring of real-world patient function and disease progression by wearing a device on the limb of interest. Second, the proposed metrics demonstrated strong clinical validity exhibited by high correlation with the total ALSFRS-RSE scores (Q1–12). Furthermore, individual question responses were associated with specific movements. Daily counts of vertical movements (flexions and extensions) demonstrated

statistically significant association with self-reported ability to dress and perform hygiene (Q6) and turning in bed and adjusting bed clothes (Q7), while daily counts of upper limb rotations (pronations and supinations) were associated with handwriting ability (Q4). Additionally, daily counts of upper limb pronations were significantly associated with cutting food and handling utensils (Q5). In addition to their association with Q1–12, all daily duration metrics were significantly associated with Q6 and Q7, but not with Q4 and Q5. The fact that duration metrics do not correlate exactly suggests that our metrics contain novel information about disease progression. For example, they are predictive of more granular behavior. Moreover, in most cases the duration metrics retained their statistical significance after adjusting for TAC, which indicates that there is a significant difference in the predicted value for each one-unit difference in limb movement duration metric when TAC remains constant. In other words, the proposed limb movement duration metrics contain significant information in estimating ALSFRS-RSE scores, even beyond the established methods for quantifying physical activity volume from wrist-worn accelerometers.²² To demonstrate our results are robust, we considered various angle thresholds and we reported results for all tests that were performed.

Third, our approach does not require accurate device positioning on the wrist, as it detects upper limb movements based on the changes of the sensor orientation angle rather than changes compared to the presumed sensor orientation. By identifying periods when motion is monotonically increasing or decreasing, our method is robust to parameters such as sampling frequency, measurement range, and amplitude resolution, and it could be applied to data collected with various wrist-worn devices that collect raw, sub-second level accelerometer data. Of note, higher sampling frequency allows one to estimate movement duration with higher precision. Also, compared to the published multi-device and multi-sensor approaches and analyses,^{42,43} the method recognizes upper limb movements from a single wrist-worn device which limits patient burden and may improve wear compliance in a free-living environment.^{33,44} As a result, our approach could allow for comparison of metrics across other ALS studies, deeper understanding of upper limb functioning across various health conditions, and establishment of upper limb movement baselines.

Our study has some limitations. First, we did not prospectively collect information on patient handedness, though we retrospectively collected this from 17 out of 20 participants. In the final sample, we included observations from the non-dominant wrist (15 left, 2 right) for participants with known handedness, and from the left wrist for the remaining three. Second, our analytical validation did not include a wide range of everyday

activities and did not employ instrumentation capable of capturing precise angular range and duration of motion. Although our main investigation focused on free-living data, quantification of movements during clinical visits (e.g., using upper limb movement trajectories such as those displayed in Fig. 1e) could improve our understanding of upper limb functioning at various stages of the disease. Further studies will need to determine the precise accuracy of the proposed metrics and optimize the proposed candidate angles (45, 90, and 135°) and number of observations for estimating the average (10).²² Nonetheless, the metrics appear to reliably track disease progression. Further studies are needed to determine the utility of the proposed metrics for comparison with other remote digital assessment tools and other patient reported outcome measures, ideally in a larger cohort that would allow for stratification of ALS phenotypes. Third, we could not differentiate movements performed with and without assistance. Moreover, our method would be unable to differentiate upper limb flexion and extension when the arm is parallel to the ground (in this scenario, during flexion pitch is decreasing, while during extension—increasing). To account for these discrepancies (flexion vs. extension, supination vs. pronation), our analysis also included aggregated metrics of $X_{fe,z}$ and $X_{sp,z}$. Finally, although our approach demonstrated a clear decline in upper limb movement, the estimated flexions and extensions assume that the angle between forearm and shoulder is constant, while in fact they are a mixture of motions originating from the forearm, shoulder, and torso. For more accurate distinction between movements of individual upper limb, the analysis should include at least two devices (one on the forearm and one on the shoulder), ideally equipped with accelerometer, gyroscope, and magnetometer (unavailable in the devices used in our study).

The presented approach offers quantifying disease progression in people with ALS through analysis of distinct upper body movements in the real-world setting. Moreover, it is accomplished using passively collected data from a single wrist-worn device. The introduced accelerometry-derived metrics closely tracked with ALS disease progression over time and metrics such as these could serve as outcome measures in ALS clinical trials, and possibly other neurological diseases affecting arm function. Finally, we believe that these measures could be combined with quantitative measures of other domains involved in the disease, such as gait, speech, respiration, and cognition, to provide a more comprehensive assessment of disease progression.

Contributors

MS—Study concept and design, method development, data processing, figure preparation, contribution to statistical data analysis, manuscript drafting.

MK—Concept and implementation of statistical data analysis, manuscript drafting.

SAJ—Study concept and design, data collection, data processing, critical review of the manuscript draft.

KMB—Study concept and design, data collection, critical review of the manuscript draft.

ZS—Study concept and design, data collection, critical review of manuscript draft.

TBR—Data collection, critical review of manuscript draft.

NC—Data collection, critical review of manuscript draft.

AC—Data collection, critical review of manuscript draft.

AI—Data collection, critical review of manuscript draft.

JB—Study concept and design, data collection, data analysis, project supervision, critical review of the manuscript.

JPO—Study concept and design, data collection, data analysis, critical review of manuscript, scientific supervision.

MS and MK have verified the underlying data.

All authors reviewed the manuscript.

All authors read and approved the final version of the manuscript.

Data sharing statement

Data may be shared upon request and after review and approval by the owners of the data. Shared data will consider deidentified summary statistics used for analytical analysis. Related correspondence should be sent to senior authors: Dr. James D. Berry (jdberry@mgm.harvard.edu) and Dr. Jukka-Pekka Onnela (onnela@hsph.harvard.edu).

Survey data were collected using Beiwie.²⁴ The calculation of forearm movement metrics was carried in MATLAB (R2022a; MathWorks, Natick, Massachusetts). All statistical analyses were performed using R software (The R Project; version 4.1.2). Both MATLAB and R codes are publicly available on GitHub repository (<https://github.com/onnela-lab/als-forearm-movements>).

Declaration of interests

SAJ—reports funding support from the ALS Association.

KMB—reports consulting fees from Cytokinetics, Inc.

JDB—reports consulting fees for advisory panels from Biogen, Clene Nanomedicine, MT Pharma of America, MT Pharma Holdings of America, Janssen, Alexion, Amylyx, Regeneron; research support and grants from Biogen, Clene Nanomedicine, MT Pharma of America, MT Pharma Holdings of America, Alexion, Amylyx, Rapa Therapeutics, Muscular Dystrophy Association, ALS Association, ALS Finding A Cure, ALS One, Tambourine, NINDS; and is on advisory panels for ALS One, Everything ALS, and MDA MOVR Database.

The other authors report no conflicts of interests.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.ebiom.2024.105036>.

References

- Goutman SA, Hardiman O, Al-Chalabi A, et al. Recent advances in the diagnosis and prognosis of amyotrophic lateral sclerosis. *Lancet Neurol*. 2022;21:480–493.
- Talbot K. Motor neuron disease: the bare essentials. *Pract Neurol*. 2009;9:303–309.
- Phukan J, Elamin M, Bede P, et al. The syndrome of cognitive impairment in amyotrophic lateral sclerosis: a population-based study. *J Neurol Neurosurg Psychiatry*. 2012;83:102–108.
- Kiernan MC, Vucic S, Cheah BC, et al. Amyotrophic lateral sclerosis. *Lancet*. 2011;377:942–955.
- Fournier CN, Bedlack R, Quinn C, et al. Development and validation of the Rasch-built overall amyotrophic lateral sclerosis disability scale (ROADS). *JAMA Neurol*. 2020;77:480–488.
- Franchignoni F, Mora G, Giordano A, Volanti P, Chiò A. Evidence of multidimensionality in the ALSFRS-R Scale: a critical appraisal on its measurement properties using Rasch analysis. *J Neurol Neurosurg Psychiatry*. 2013;84:1340–1345.
- Montes J, Levy G, Albert S, et al. Development and evaluation of a self-administered version of the ALSFRS-R. *Neurology*. 2006;67:1294–1296.
- Richards D, Morren JA, Pioro EP. Time to diagnosis and factors affecting diagnostic delay in amyotrophic lateral sclerosis. *J Neurol Sci*. 2020;417:117054.
- Deodhar A, Gensler LS, Magrey M, et al. Assessing physical activity and sleep in axial spondyloarthritis: measuring the gap. *Rheumatol Ther*. 2019;6:487–501.
- Nikbakhtian S, Reed AB, Obika BD, et al. Accelerometer-derived sleep onset timing and cardiovascular disease incidence: a UK Biobank cohort study. *Eur Heart J Digit Health*. 2021;2:658–666.
- Lämsä M, Kangas M, Jokelainen J, et al. Cardiovascular disease risk and all-cause mortality associated with accelerometer-measured physical activity and sedentary time – a prospective population-based study in older adults. *BMC Geriatr*. 2022;22:729.
- Ezeugwu V, Klaren RE, A. Hubbard E, Manns PT, Motl RW. Mobility disability and the pattern of accelerometer-derived sedentary and physical activity behaviors in people with multiple sclerosis. *Prev Med Rep*. 2015;2:241–246.
- Garcia-Gancedo L, Kelly ML, Lavrov A, et al. Objectively monitoring amyotrophic lateral sclerosis patient symptoms during clinical trials with sensors: observational study. *JMIR Mhealth Uhealth*. 2019;7:e13433.
- van Eijk RPA, Bakers JNE, Bunte TM, de Fockert AJ, Eijkemans MJC, van den Berg LH. Accelerometry for remote monitoring of physical activity in amyotrophic lateral sclerosis: a longitudinal cohort study. *J Neurol*. 2019;266:2387–2395.
- Beukenhorst AL, Collins E, Burke KM, et al. Smartphone data during the COVID-19 pandemic can quantify behavioral changes in people with ALS. *Muscle Nerve*. 2021;63:258–262.
- Berry JD, Paganoni S, Carlson K, et al. Design and results of a smartphone-based digital phenotyping study to quantify ALS progression. *Ann Clin Transl Neurol*. 2019;6:873–881.
- Holdom CJ, van Unnik JWJ, van Eijk RPA, et al. Use of hip- versus wrist-based actigraphy for assessing functional decline and disease progression in patients with motor neuron disease. *J Neurol*. 2023;270:2597–2605.
- Vieira FG, Venugopalan S, Premasiri AS, et al. A machine-learning based objective measure for ALS disease severity. *NPJ Digit Med*. 2022;5:45.
- Ratti E, Berry J, Vangel M, Macklin E, Schoenfeld D, Cudkowicz M. Progression to clinically meaningful changes in ALSFRS-R bulbar and fine motor domains is faster in bulbar onset and in limb onset amyotrophic lateral sclerosis patients respectively (P5.051). *Neurology*. 2015;84:P5.051.
- Dieu O, Mikulovic J, Fardy PS, Bui-Xuan G, Beghin L, Vanhelst J. Physical activity using wrist-worn accelerometers: comparison of dominant and non-dominant wrist. *Clin Physiol Funct Imaging*. 2017;37:525–529.
- Johnson SA, Karas M, Burke KM, et al. Wearable device and smartphone data quantify ALS progression and may provide novel outcome measures. *NPJ Digit Med*. 2023;6:34.
- Varma VR, Dey D, Leroux A, et al. Total volume of physical activity: TAC, TLAC or TAC(λ). *Prev Med*. 2018;106:233–235.
- Brooks BR. El Escorial world federation of neurology criteria for the diagnosis of amyotrophic lateral sclerosis. Subcommittee on motor neuron diseases/amyotrophic lateral sclerosis of the world federation of neurology research group on neuromuscular diseases and the El Escorial “Clinical limits of amyotrophic lateral sclerosis” workshop contributors. *J Neurol Sci*. 1994;124(Suppl):96–107.
- Onnela J-P, Dixon C, Griffin K, et al. Beiwie: a data collection platform for high-throughput digital phenotyping. *J Open Source Softw*. 2021;6:3417.
- van Hees VT, Gorzelniak L, Dean León EC, et al. Separating movement and gravity components in an acceleration signal and implications for the assessment of human daily physical activity. *PLoS One*. 2013;8:e61691.
- Pedley M. Tilt sensing using a three-axis accelerometer. *Free Semicond Appl Note*. 2013;1:2012–2013.

- 27 Choi L, Liu Z, Matthews CE, Buchowski MS. Validation of accelerometer wear and nonwear time classification algorithm. *Med Sci Sports Exerc.* 2011;43:357–364.
- 28 ALS functional rating scale revised (ALS-FRS-R). <https://www.encals.eu/wp-content/uploads/2016/09/ALS-Functional-Rating-Scale-Revised-fill-in-form.pdf>.
- 29 Neishabouri A, Nguyen J, Samuelsson J, et al. Quantification of acceleration as activity counts in ActiGraph wearable. *Sci Rep.* 2022;12:11958.
- 30 Nakagawa S, Schielzeth H. A general and simple method for obtaining R² from generalized linear mixed-effects models. *Methods Ecol Evol.* 2013;4:133–142.
- 31 Johnson PCD. Extension of Nakagawa & Schielzeth's R²GLMM to random slopes models. *Methods Ecol Evol.* 2014;5:944–946.
- 32 Mukaka MM. Statistics corner: a guide to appropriate use of correlation coefficient in medical research. *Malawi Med J.* 2012;24:69–71.
- 33 Hayden CD, Murphy BP, Hardiman O, Murray D. Measurement of upper limb function in ALS: a structured review of current methods and future directions. *J Neurol.* 2022;269:4089–4101.
- 34 Biswas D, Corda D, Baldus G, et al. Recognition of elementary arm movements using orientation of a tri-axial accelerometer located near the wrist. *Physiol Meas.* 2014;35:1751–1768.
- 35 Biswas D, Cranny A, Gupta N, et al. Recognizing upper limb movements with wrist worn inertial sensors using k-means clustering classification. *Hum Mov Sci.* 2015;40:59–76.
- 36 Gupta AS, Luddy AC, Khan NC, Reiling S, Thornton JK. Real-life wrist movement patterns capture motor impairment in individuals with ataxia-telangiectasia. *Cerebellum.* 2022;22:1–11.
- 37 Franchignoni F, Mandrioli J, Giordano A, Ferro S. A further Rasch study confirms that ALSFRS-R does not conform to fundamental measurement requirements. *Amyotroph Lateral Scler Frontotemporal Degener.* 2015;16:331–337.
- 38 Pinto S, Gromicho M, de Carvalho M. Assessing upper limb function with ALSFRS-R in amyotrophic lateral sclerosis patients. *Amyotroph Lateral Scler Frontotemporal Degener.* 2019;20:445–448.
- 39 Grandez K, Solas G, Bustamante P, Sedano B. Sensor device for testing activities in Parkinson and ALS patients. In: *2010 4th international conference on pervasive computing technologies for health-care.* 2010:1–6.
- 40 Gupta AS, Patel S, Premasiri A, Vieira F. At-home wearables and machine learning sensitively capture disease progression in amyotrophic lateral sclerosis. *Nat Commun.* 2023;14:5080.
- 41 Londral A, Pinto S, de Carvalho M. Markers for upper limb dysfunction in Amyotrophic Lateral Sclerosis using analysis of typing activity. *Clin Neurophysiol.* 2016;127:925–931.
- 42 Mazomenos EB, Biswas D, Cranny A, et al. Detecting elementary arm movements by tracking upper limb joint angles with MARG sensors. *IEEE J Biomed Health Inform.* 2016;20:1088–1099.
- 43 Smith BA, Lang CE. Sensor measures of symmetry quantify upper limb movement in the natural environment across the lifespan. *Arch Phys Med Rehabil.* 2019;100:1176–1183.
- 44 Troiano RP, McClain JJ, Brychta RJ, Chen KY. Evolution of accelerometer methods for physical activity research. *Br J Sports Med.* 2014;48:1019–1023.